

Systematic review

1. * Review title.

Give the working title of the review, for example the one used for obtaining funding. Ideally the title should state succinctly the interventions or exposures being reviewed and the associated health or social problems. Where appropriate, the title should use the PI(E)COS structure to contain information on the Participants, Intervention (or Exposure) and Comparison groups, the Outcomes to be measured and Study designs to be included.

INterventions for Complex Traumatic Events – INCiTE

2. Original language title.

For reviews in languages other than English, this field should be used to enter the title in the language of the review. This will be displayed together with the English language title.

3. * Anticipated or actual start date.

Give the date when the systematic review commenced, or is expected to commence.

01/03/2017

4. * Anticipated completion date.

Give the date by which the review is expected to be completed.

31/08/2018

5. * Stage of review at time of this submission.

Indicate the stage of progress of the review by ticking the relevant Started and Completed boxes. Additional information may be added in the free text box provided.

Please note: Reviews that have progressed beyond the point of completing data extraction at the time of initial registration are not eligible for inclusion in PROSPERO. Should evidence of incorrect status and/or completion date being supplied at the time of submission come to light, the content of the PROSPERO record will be removed leaving only the title and named contact details and a statement that inaccuracies in the stage of the review date had been identified.

This field should be updated when any amendments are made to a published record and on completion and publication of the review. If this field was pre-populated from the initial screening questions then you are not able to edit it until the record is published.

The review has not yet started: No

Review stage	Started	Completed
Preliminary searches	Yes	No
Piloting of the study selection process	No	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Provide any other relevant information about the stage of the review here (e.g. Funded proposal, protocol not yet finalised).

6. * Named contact.

The named contact acts as the guarantor for the accuracy of the information presented in the register record.

Peter Coventry

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

7. * Named contact email.

Give the electronic mail address of the named contact.

peter.coventry@york.ac.uk

8. Named contact address

Give the full postal address for the named contact.

Mental Health & Addition Research Group

Department of Health Sciences

Faculty of Science

Area 4, Room A/RC/205, ARRC Building

University of York

York

YO10 5DD

9. Named contact phone number.

Give the telephone number for the named contact, including international dialling code.

+44(0)1904 321528

10. * Organisational affiliation of the review.

Full title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

University of York

Organisation web address:

<https://www.york.ac.uk/>

11. * Review team members and their organisational affiliations.

Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. **NOTE: email and country are now mandatory fields for each person.**

Dr Peter Coventry. Department of Health Sciences and Centre for Reviews and Dissemination, University of York, UK

Professor Rachel Churchill. Centre for Reviews and Dissemination, University of York and Coordinating Editor, Cochrane Common Mental Health Disorders Group, UK

Professor Simon Gilbody. Department of Health Sciences, University of York, UK

Professor Karina Lovell. School of Nursing, Midwifery and Social Work, University of Manchester, UK

Professor Corrado Barbui. Section of Psychiatry, University of Verona, Italy

Dr Nick Meader. Centre for Reviews and Dissemination, University of York, UK

Dr Dean McMillan. Department of Health Sciences, University of York, UK

Ms Kath Wright. Centre for Reviews and Dissemination, University of York, UK

Dr Melanie Temple. The Retreat, York

Dr Jane Dalton. Centre for Reviews and Dissemination, University of York, UK

Ms Hollie Melton. Centre for Reviews and Dissemination, University of York

12. * Funding sources/sponsors.

Give details of the individuals, organizations, groups or other legal entities who take responsibility for initiating, managing, sponsoring and/or financing the review. Include any unique identification numbers assigned to the review by the individuals or bodies listed.

NIHR HTA Programme (Project ref 16/11)

Grant number(s)

13. * Conflicts of interest.

List any conditions that could lead to actual or perceived undue influence on judgements concerning the main topic investigated in the review.

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country are now mandatory fields for each person.**

Professor Jonathan Bisson. Cardiff University, UK

Ms Stephanie Wetherhill. The Tuke Centre, York, UK

Ms Lizzy Ferguson. The Retreat, York, UK

Professor Marylene Cloitre. New York University, USA

Professor Thanos Karatzias. Napier University, Edinburgh, UK

Dr Neil Roberts. Cardiff University, UK

15. * Review question.

State the question(s) to be addressed by the review, clearly and precisely. Review questions may be specific

or broad. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS where relevant.

What randomised and non-randomised controlled trials exist of psychological and/or pharmacological interventions for mental health in people with a history of complex traumatic events, with specific reference to participant characteristics, intervention format and content, and outcomes measured?

What is the clinical effectiveness of interventions delivered to adults aged 18 and over with trauma and stressor disorders after exposure to complex traumatic events?

What is the comparative effectiveness of psychological interventions and different pharmacological interventions

What are service user and provider perspectives on the acceptability and feasibility of using psychological and/or pharmacological interventions to treat mental health problems after complex traumatic events, with specific reference to professional competencies and training, intervention uptake, adherence, and service experience.

What candidate interventions could be feasibly tested and used in the NHS and what future research priorities might be recommended to the HTA programme?

16. * Searches.

State the sources that will be searched. Give the search dates, and any restrictions (e.g. language or publication period). Do NOT enter the full search strategy (it may be provided as a link or attachment.)

For the effectiveness synthesis, we will include electronic database searches (e.g. MEDLINE, Embase, PsycINFO, CINAHL, Science Citation Index as well as specialised databases such as International Pharmaceutical Abstracts and PILOTS – Published International Literature on Traumatic Stress), journal hand searches, reference list searches, targeted author searches, grey literature searches (including material generated by service user-led organisations), research register searches, and forward citation searching. No language or other restrictions will apply. In partnership with our Advisory Group we will make judgements about the relevance of international literature to the UK health setting against our stated inclusion criteria.

For the qualitative acceptability synthesis, we will run a separate search to identify qualitative evidence that addresses our aim to explore service user experience of treatments for people with a history of complex traumatic events. Search terms relating to qualitative designs will be developed based on our previous meta-syntheses and terms relating to user experience of therapies will be developed by scanning the background literature, browsing the MEDLINE medical subject heading (MeSH) thesaurus and via discussion between the research team. Additionally, while screening effectiveness studies for eligibility we will flag qualitative process evaluations that are associated with clinical trials.

17. URL to search strategy.

Give a link to a published pdf/word document detailing either the search strategy or an example of a search

strategy for a specific database if available (including the keywords that will be used in the search strategies), or upload your search strategy. Do NOT provide links to your search results.

Alternatively, upload your search strategy to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

Yes I give permission for this file to be made publicly available

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied. This could include health and wellbeing outcomes.

Mental health problems associated with a history of complex traumatic events.

19. * Participants/population.

Give summary criteria for the participants or populations being studied by the review. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion: Adults aged 18 or over exposed to complex interpersonal traumatic events. This will include adults exposed to childhood physical and/or sexual abuse, being a victim of or witnessing domestic violence, forcibly displaced persons (refugees, asylum seekers, internally displaced persons), torture survivors, recruitment into armed conflict as a child, on-going armed conflict and combat, and relocation through human trafficking.

Exclusion: Participants with a history of exposure to single and non-interpersonal traumatic events (such as a road traffic incident or a natural disaster).

We anticipate that some studies will include a mix of participants with a history of single event non-interpersonal trauma and complex interpersonal trauma. We will therefore include studies of mixed populations if we can determine that 75% of the participants have a history of complex trauma. If studies include less than 75% of participants with complex trauma but present data separately for this sub-group we will consider inclusion if these types of mixed population studies meet the other PICO criteria.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the nature of the interventions or the exposures to be reviewed.

Psychological interventions: This review will include studies that evaluated any first or second line psychological therapy aimed at improving symptoms (including comorbidities) of trauma- and stressor related disorders either delivered to individuals or in a group. In-keeping with the classification used by NICE, interventions to be considered are:

- i. Trauma-focused CBT that includes one or more of the following types of treatment techniques: exposure, cognitive therapy, stress management.
- ii. Eye movement desensitisation and reprocessing.
- iii. Other psychological treatments used to treat trauma survivors and victims but use predominately non-CBT techniques: supportive therapy and non-directive counselling; psychodynamic therapies, including inter

personal psychotherapy (IPT); hypnotherapy; mindfulness and compassion focused therapies; acceptance and commitment therapies; accelerated resolution, and sensorimotor therapies.

Where possible we will include as part of sub-group analyses consideration of the following interventions:

iv. Group trauma-focused CBT

v. Group non-trauma focused CBT

Pharmacological interventions: In the UK only two drugs are licensed for the treatment of PTSD: paroxetine and (in women only) sertraline. However this review will consider all drug treatments subjected to experimental testing in the context of treatment of mental health problems in people with a history of complex trauma. Categories of pharmacotherapy to be considered are: SSRIs, SNRIs, tricyclic antidepressants, anxiolytic medication, mood stabilizers, atypical antipsychotics.

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the main subject/topic of the review will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

Psychological interventions versus

i. Waitlist

ii. Treatment as usual

iii. Symptom monitoring

iv. Repeated assessment or other minimal attention control group akin to psychological placebo

v. Alternative psychological treatment

vi. Pharmacological treatment

Pharmacological interventions versus

i. Placebo

ii. Other medication

iii. Psychological therapy

Comparisons of two or more active interventions or of an active treatment with a 'no treatment' comparator will be included. Differences in comparators will be taken into account during data summary and analyses.

Network meta-analyses will be conducted to provide comparisons of all interventions within a connected network (including comparisons of active interventions not originally evaluated in included trials).

22. * Types of study to be included.

Give details of the types of study (study designs) eligible for inclusion in the review. If there are no restrictions on the types of study design eligible for inclusion, or certain study types are excluded, this should be stated. The preferred format includes details of both inclusion and exclusion criteria.

Experimental studies (effectiveness synthesis): We will include randomised controlled trials (RCTs) and cluster RCTs (if relevant; non-randomised controlled studies will be included to capture data on emerging

treatments and treatments tested in more pragmatic settings. Single group before and after studies, uncontrolled observational studies, case studies, opinion papers, descriptive studies, editorials will be excluded. Qualitative studies (acceptability and feasibility synthesis): For the purpose of our review, qualitative research will be defined as those studies that collect data using specific qualitative techniques such as unstructured interviews, semi-structured interviews or focus groups, either as a stand-alone methodology or as discrete part of a larger mixed-method study, and analysed qualitatively. Studies that have collected data using qualitative methods but then analysed these data using quantitative methods will therefore be excluded.

23. Context.

Give summary details of the setting and other relevant characteristics which help define the inclusion or exclusion criteria.

Studies undertaken in any country and setting (i.e. LMIC and HIC) are eligible for inclusion across all phases of the review.

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

- i. Reduction in severity of traumatic stress symptoms as measured using a validated and standardised clinician rated scale.
- ii. Reduction in symptoms of difficulties with emotion regulation (e.g. Difficulties with Emotion Regulation Scale [DERS]) and interpersonal relationship problems (e.g. Inventory of Interpersonal Problems [IIP]).

* Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

- i.. Severity of self-reported traumatic stress symptoms using a standardised measure (e.g. Modified PTSD Symptom Scale (MPSS-SR)
- ii. Reduction in depressive and/or anxiety symptoms measured using validated clinician-rated instruments (e.g. Hamilton Depression Rating Scale), or validated patient self-reported instruments (e.g. Hospital Anxiety and Depression Scale)
- iii. Reduction in symptoms of panic disorder
- iv. Reduction in symptoms of disassociation

- v. Reduction in symptoms of functional somatic syndromes
- vi. Reduction in substance misuse
- vii. Acceptability measured in terms of intervention uptake, adherence and withdrawal (drop-outs)
- viii. Adverse events and harms from trial data (e.g. worsening of traumatic stress symptoms); for psychological interventions we will code these in-keeping with latest typologies of adverse events of psychological therapies.
- ix. Suicidal ideation, attempts, and completion
- x. Functioning and disability and quality of life measured by validated clinician-rated scales (e.g. Global Assessment of Functioning) or validated self-reported scales (e.g. SF-36, Sheehan Disability Scale).

* Measures of effect

Please specify the effect measure(s) for you additional outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat'.

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

Two reviewers will independently screen titles and abstracts for relevance against the inclusion criteria.

Where both reviewers agree on inclusion, or where there is disagreement, the full text article will be retrieved. The two reviewers will independently assess the full text of the articles against the inclusion criteria. Any remaining disagreements will be resolved through discussion with other members of the team.

We will record inclusions and exclusions throughout this process to enable us to complete a PRISMA flow chart and produce tables of 'Characteristics of included studies' and 'Characteristics of excluded studies'.

Data extraction will be performed by one reviewer and independently checked="checked" value="1" by a second. Discrepancies will be resolved by referral to the original studies and if necessary through arbitration by a third reviewer. Data extraction will be guided by a pre-specified data extraction sheet detailing key features of the study sample, setting, methods, intervention, control (if appropriate) and recorded and reported outcome measures. Short-term (post treatment to six months), long-term (7-12 months) and very long-term data (12 months) will be extracted and analysed separately.

27. * Risk of bias (quality) assessment.

Describe the method of assessing risk of bias or quality assessment. State which characteristics of the studies will be assessed and any formal risk of bias tools that will be used.

Quantitative (effectiveness) synthesis: Included RCTs will be assessed for risk of bias using the Cochrane Collaboration Risk of Bias (RoB) assessment tool for randomised controlled trials. We will assess risk of bias in non-randomised studies using the Cochrane Risk of Bias for Non-Randomised Studies of Interventions (ROBINS-I).

Qualitative (acceptability) synthesis: We will adopt the CerQual (certainty of the qualitative evidence) approach to assess both the methodological limitations of individual studies and the coherence of our review findings. Methodological limitations will be assessed with the CASP checklist.

Risk of bias (quality) assessment will be performed by one reviewer and independently checked="checked" value="1" by a second. Discrepancies will be resolved by referral to the original studies and if necessary through arbitration by a third reviewer.

28. * Strategy for data synthesis.

Provide details of the planned synthesis including a rationale for the methods selected. This **must not be generic text** but should be **specific to your review** and describe how the proposed analysis will be applied to your data.

Clinical effectiveness of psychological and/or pharmacological interventions versus control:

We will conduct meta-analyses pooling data relating to psychological and/or pharmacological interventions where the interventions, populations and study contexts are sufficiently similar to make such analyses appropriate and interpretable. We will explore statistical heterogeneity thoroughly through use of appropriate statistics such as I-squared. If cluster trials are identified, they will be analysed appropriately, which may involve reducing effective sample size through calculation of the 'design effect'. Studies including multiple treatment groups will be analysed including each relevant pair-wise comparison separately, but with shared intervention groups divided out approximately evenly among the comparisons to avoid double counting.

Where studies report multiple outcomes within one of our categories (e.g. two depression scores), we will use one, based on a decision rule that maximises comparability between studies in the review (i.e. including measures which are used frequently by other studies in the review) or take a median in cases where the decision rule does not apply. We will apply measures of effect (such as the standardised mean difference) so that the results of different interventions can be compared by decision-makers to assess their relative value.

In recognition that the outputs of reviews need to be accessible and relevant to wider range of audiences including service users and professionals, we will, where certain assumptions for the variance of effect are met, back-transform standardised effects into mean differences, and to translate standardised effects into clinical meaningful metrics such as number needed to treat.

Comparative effectiveness of psychological interventions – network meta-analysis:

Where data permits our proposed network meta analyses will allow the full consideration of the relative effectiveness of all psychological interventions, including those that have not been directly compared.

The structure of the network meta-analyses will be informed by clinical considerations, ensuring that the pooled estimates are clinically meaningful. In the absence of standardisation of conduct and reporting of mixed treatment comparisons, we will follow the good research practices detailed in the International Society for Pharmacoeconomics and Outcomes Research (ISPOR) Task Force guidance. We will use methods developed and illustrated in NICE Technical Support Document 2, which details the generalized linear

modelling framework for network meta-analyses of randomized controlled trials. Estimates of treatment effects will be calculated using a Bayesian approach (using freely available software: WinBUGS), taking into account the correlation between multi-arm trials where appropriate. Direct and indirect treatment effects will be estimated, along with credible intervals and prediction intervals. The network meta-analysis method enables the comparison of indirect effects, treatment comparisons not addressed within the primary trials. Such network analysis can only be applied to connected networks of trials. We will evaluate model diagnostics and assumptions of similarity and consistency in the evidence network. As it is common to simultaneously consider the effects of an intervention whilst also considering for example acceptability, we will present the relative ranking for these two outcomes simultaneously.

Where sufficient studies allow, we will explore the possibility of estimation and adjustment of bias using the risk of bias domains random sequence generation, allocation concealment, and blinding (as per the meta-analysis synthesis). To aid readability we will use graphical methods to illustrate the comparative effectiveness of the different interventions.

Qualitative synthesis: We will use a narrative synthesis approach. Consistent with previous NIHR funded qualitative syntheses that we have conducted, we will place greater emphasis on studies presenting 'thicker' data descriptions (i.e. those studies employing rigorous data collection/analysis techniques and providing in-depth examinations of user experience). Thinner data descriptions (i.e. those which lack this detail) will be used to augment and contextualise the findings. The relative contribution of individual studies, the impact of methodological limitations, and certainty on the findings will be summarised narratively in line with the CerQual approach.

Once we have extracted and synthesised all available quantitative and qualitative data, we will seek, wherever possible, to integrate our findings. This will be achieved by exploring the extent to which our identified qualitative themes map onto the quantitative acceptability ratings about intervention uptake.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

To answer whether treatments are more effective than other treatments for specific population sub-groups we will perform subgroup meta-analyses for the primary and secondary outcomes of critical importance to mental health. The population subgroups will include victims of specific types of trauma:

- Exposure to sexual abuse as a child
- Exposure to sexual abuse as an adult
- Refugees and asylum seekers
- Trafficking and exploitation.

30. * Type and method of review.

Select the type of review and the review method from the lists below. Select the health area(s) of interest for your review.

Type of review

Cost effectiveness

No

Diagnostic

No

Epidemiologic

No

Individual patient data (IPD) meta-analysis

No

Intervention

No

Meta-analysis

No

Methodology

No

Narrative synthesis

No

Network meta-analysis

No

Pre-clinical

No

Prevention

No

Prognostic

No

Prospective meta-analysis (PMA)

No

Review of reviews

No

Service delivery

No

Synthesis of qualitative studies

No

Systematic review

Yes

Other

No

Health area of the review

Alcohol/substance misuse/abuse

No

Blood and immune system

No

Cancer

No

Cardiovascular

No

Care of the elderly

No

Child health

No

Complementary therapies

No

Crime and justice

No

Dental

No

Digestive system

No

Ear, nose and throat

No

Education

No

Endocrine and metabolic disorders

No

Eye disorders

No

General interest

No

Genetics

No

Health inequalities/health equity

No

Infections and infestations

No

International development

No

Mental health and behavioural conditions

No

Musculoskeletal

No

Neurological

No

Nursing

No

Obstetrics and gynaecology

No

Oral health

No

Palliative care

No

Perioperative care

No

Physiotherapy

No

Pregnancy and childbirth

No

Public health (including social determinants of health)

No

Rehabilitation
No

Respiratory disorders
No

Service delivery
No

Skin disorders
No

Social care
No

Surgery
No

Tropical Medicine
No

Urological
No

Wounds, injuries and accidents
No

Violence and abuse
No

31. Language.

Select each language individually to add it to the list below, use the bin icon to remove any added in error.
English

There is an English language summary.

32. * Country.

Select the country in which the review is being carried out from the drop down list. For multi-national collaborations select all the countries involved.

England

33. Other registration details.

Give the name of any organisation where the systematic review title or protocol is registered (such as with The Campbell Collaboration, or The Joanna Briggs Institute) together with any unique identification number assigned. (N.B. Registration details for Cochrane protocols will be automatically entered). If extracted data will be stored and made available through a repository such as the Systematic Review Data Repository (SRDR), details and a link should be included here. If none, leave blank.

34. Reference and/or URL for published protocol.

Give the citation and link for the published protocol, if there is one

Give the link to the published protocol.

https://www.crd.york.ac.uk/PROSPEROFILES/55523_PROTOCOL_20170320.pdf

Alternatively, upload your published protocol to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

Yes I give permission for this file to be made publicly available

Please note that the information required in the PROSPERO registration form must be completed in full even

if access to a protocol is given.

35. Dissemination plans.

Give brief details of plans for communicating essential messages from the review to the appropriate audiences.

The review will be reported according to PRISMA guidelines and submitted in a final report to the NIHR HTA programme, for publication as a monograph in the NIHR Journals Library. A series of publications describing different aspects of the project (e.g. clinical effectiveness, acceptability) will be written and submitted to high impact academic and practice journals. We will identify existing Cochrane reviews that might be updated or identify opportunities for new Cochrane reviews to address additional unanswered questions about treating and measuring outcomes in people with complex trauma histories. Abstracts will be submitted to relevant major national and international conferences such as the Royal College of Psychiatrists International Congress. Presentations to professional, research, and service user audiences will also be made at key events organised by partner organisations such as NIHR CLAHRC Yorkshire and Humber which hosts a mental health and comorbidity theme, and the Northern IAPT Practice Research Network which hosts two annual events to showcase research relevant to delivery of low and high intensity psychological services. Dissemination will also be supported by a pro-active social media campaign to promote the programme among professional and public audiences, report on progress and highlight key achievements, and cite more broadly research and patient initiatives and trends in treating people with complex trauma histories. Social media communication will include a bespoke Twitter account for INCiTE, and the programme will be more widely promoted among professionals, charities, services and service users and the research community via institutional Twitter accounts that are followed by more than 106K (@MHARG_york; @crd_york; @TheTukeCentre, @TheYorkMind; @HealthSciYork; @CLAHRCYH; @NIHR_DC; @OfficialNIHR; @cochraneollab; @Cochrane_CCMD).

Do you intend to publish the review on completion?

Yes

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords will help users find the review in the Register (the words do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

Complex-PTSD (CPTSD)

Posttraumatic stress disorder (PTSD)

Disorders of traumatic stress

Depression

Anxiety

Prolonged trauma

Mental health

37. Details of any existing review of the same topic by the same authors.

Give details of earlier versions of the systematic review if an update of an existing review is being registered, including full bibliographic reference if possible.

38. * Current review status.

Review status should be updated when the review is completed and when it is published. For newregistrations the review must be Ongoing.

Please provide anticipated publication date

Review_Ongoing

39. Any additional information.

Provide any other information the review team feel is relevant to the registration of the review.

40. Details of final report/publication(s).

This field should be left empty until details of the completed review are available.

Give the link to the published review.